ASSOCIATION OF MATHEMATICS TEACHRS OF INDIA

Screening Test – KAPREKAR Contest (NMTC Sub-Junior Level—VII and VIII Grades)

2025-2026

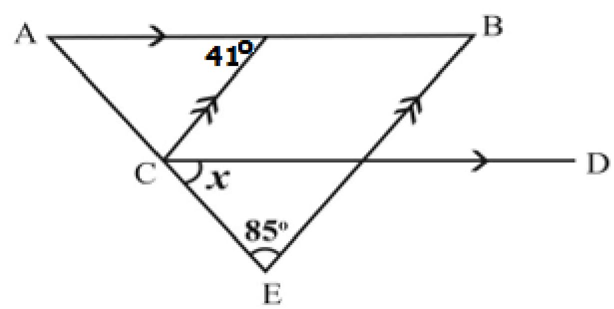
Instructions: (Read Carefully)

1. Fill in the Response sheet with your Name, Class and the Institution through which you appear, in the specified places.
2. Diagrams are only Visual guides; they are not drawn to scale.
3. You may use separate sheets to do rough work.
4. Use of Electronic gadgets such as Calculator, Mobile Phone or Computer is not permitted.
5. Duration of Test: 10 am to 12 Noon (Two hours)
6. For each correct response you get 1 mark; for each incorrect response, you lose ½ mark.

01) AB is a straight road of length 400 metres. From A, Samrud runs at a speed of $6m/s$ towards B and at the same time Saket starts from B and runs towards A at a speed of $5m/s$. After reaching their destinations, they return with the same speeds. They repeat it again and again. How many times do they meet each other in 15 minutes?

A) 25 **B)** 23 **C)** 24 **D)** 20

02) In the adjoining figure, the measure of the angle x is



A) 84° **B)** 44°

C) 64° **D)** 54^0

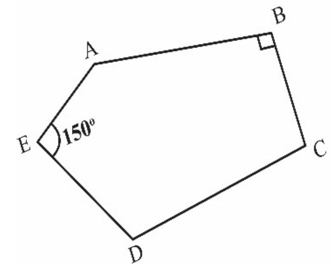
03) The value of x which satisfies $\frac{1}{x+a}+\frac{1}{x+b}=\frac{1}{x+a+b}+\frac{1}{x}$ is

A) $\frac{a+b}{2}$ **B)** $\frac{a-b}{2}$ **C)** $\frac{b-a}{2}$ **D)** $\frac{-(a+b)}{2}$

04) Two sides of an isosceles triangle are $23cm$ and $17cm$ respectively. The perimeter of the triangle (in cm) is

A) 63 **B)** 57 **C)** 63 *or* 57 **D)** 40

05) *ABCDE* is a pentagon with $\angle B = 90^{o}$ and $\angle E = 150^{o}$.
If $\angle C + \angle D = 180^{o}$ and $\angle A + \angle D = 180^{o}$, then the external angle $\angle D$ is



A) 120^{o} **B)** 110^{o}

C) 105^{o} **D)** 115^{o}

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06) The unit's digit of the product $3^{2025} \times 7^{2024}$ is

A) 1 **B)** 2 **C)** 3 **D)** 6

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07) The smallest positive integer n for which $18900 \times n$ is a perfect cube is

A) 189 **B)** 18900 **C)** 21 **D)** 490

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08) Two numbers a and b are respectively 20% and 50% more of a third number c. The percentage of a to b is

A) 120% **B)** 80% **C)** 75% **D)** 110%

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09) If $a + b = 2, \frac{1}{a} + \frac{1}{b} = 18$, then $a^3 + b^3$ lies between

A) 7 and 8 **B)** 6 and 7 **C)** 8 and 9 **D)** 5 and 6

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10) If $\sqrt{12 + \sqrt[3]{x}} = \frac{7}{2}$ and $x = \frac{p}{q}$, p, q are natural numbers with $G.C.D.(p, q) = 1$, then $p + q$ is

A) 65 **B)** 56 **C)** 45 **D)** 54

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11) The smallest number of 4-digits leaving a remainder 1 when divided by 2 or 3 or 4 or 6 has

A) 5 as its unit digit **B)** Only one zero as one of the digits
C) Exactly two zeroes as its digits **D)** 7 as its unit digit

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12) If $a : b = 2 : 3$, $b : c = 4 : 5$ and $a + c = 736$, then the value of b is

A) 392 **B)** 378 **C)** 384 **D)** 386

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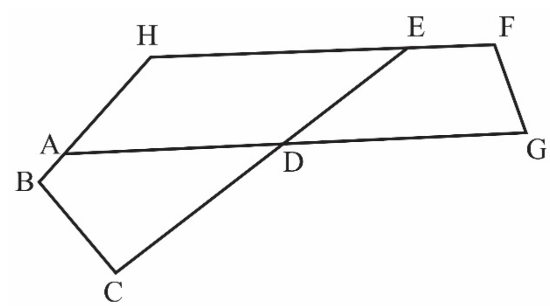
13) In the given figure,
$\angle B = 110^{o}$; $\angle C = 80^{o}$;
$\angle F = 120^{o}$; $\angle ADC = 30^{o}$
$2\angle DGF = \angle DEF$.
The measure of $\angle BHF$ is



A) 115^{o} **B)** 135^{o}

C) 100^{o} **D)** 130^{o}

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14) If $\frac{1}{b+c}+\frac{1}{c+a}=\frac{2}{a+b}$, then the value of $\frac{a^2+b^2}{c^2}$ is

A) 2 **B)** 1 **C)** ½ **D)** 3

15) If 3 men *or* 4 women can do a job in 43 days, the number of days the same job is done by 7 men and 5 women is

A) 12 **B)** 10 **C)** 11 **D)** 13

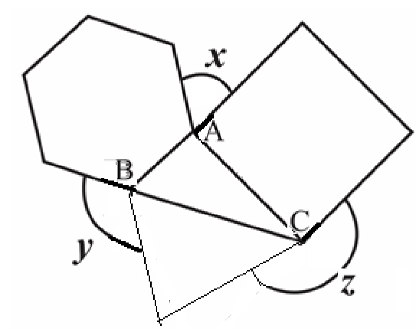
FILL IN THE BLANKS

16) The expression $49(a+b)^2-46(a-b)^2$ is factorized into $(la+mb)(na+pb)$, then the numerical value of $(l+m+n+p)$ is ________________.

17) The integer part of the solution of the equation in x,

$$\frac{1}{3}(x-3)-\frac{1}{4}(x-8)=\frac{1}{5}(x-5)$$ is ____________.

18) In the adjoining figure, *ABC* is a triangle in which $\angle BAC = 100^o$, $\angle ACB = 30^o$. An equilateral triangle, a square and a regular hexagon are drawn as shown in the figure. The measure (in degrees) of $(x + y + z)$ is ___________.

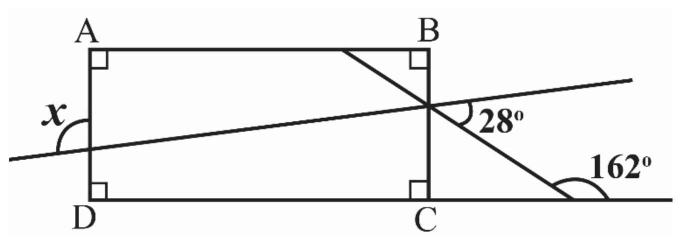


19) The mean of 5 numbers is 105. The first number is $\frac{2}{5}$ times the sum of the other 4 numbers. The first number is ____________.

20) *PQRS* is a square. The sides *PQ* and *RS* are increased by 30% each and the sides *QR* and *PS* are increased by 20% each. The area of the quadrilateral thus obtained exceeds the area of the square by _______%.

21) If $x^2+(2+\sqrt{3})x-1=0$ and $x^2+\frac{1}{x^2}=a+b\sqrt{c}$, then $(a+b+c)$ is _________.

22) In the given figure, *ABCD* is a rectangle. The measure of angle x is _______ degrees.



23) The sum of all positive integers m,n which satisfy $m^2+2mn+n=44$ is ____

24) Given a = 2025, b = 2024, the numerical value of

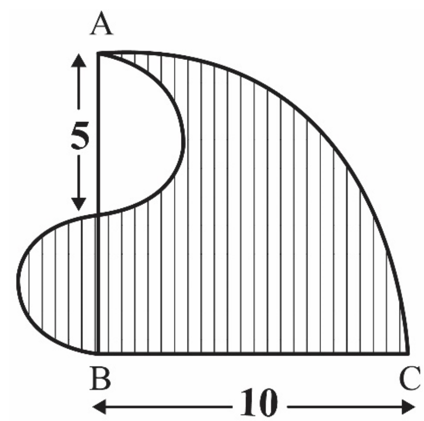
$$\left(a+b-\frac{4ab}{a+b}\right)\div\left(\frac{a}{a+b}-\frac{b}{b-a}+\frac{2ab}{b^2-a^2}\right) \text{ is } \_\_\_\_\_\_\_\_\_\_$$

25) In the sequence 0, 7, 26, 63, 124, … … … the 6th term is ________

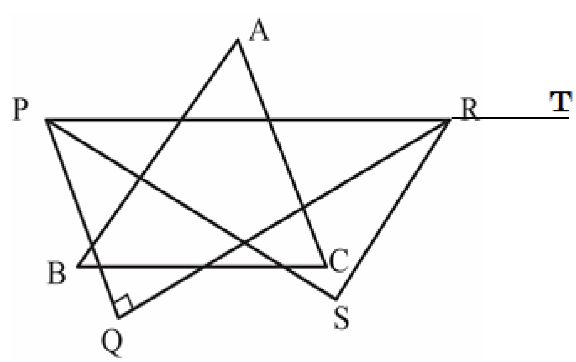
26) If $A = \sqrt{281+\sqrt{53+\sqrt{112+\sqrt{81}}}}$, $B = \sqrt{92+\sqrt{55+\sqrt{75+\sqrt{36}}}}$,
then $A - B$ is _____

27) The average of the numbers a, b, c, d is $(b + 4)$. The average of pairs (a, b), (b, c) and (c, a) are respectively 16, 26 and 25. Then the average of d and 67 is _______

28) ABC is a quadrant of a circle of radius 10 *cm*. Two semicircles are drawn as in the figure. The area of the shaded portion is $k\pi$, where k is a positive integer. The value of k is ____________



29) In the figure, ABC and PQR are two triangles such that $\angle A : \angle B : \angle C = 5 : 6 : 7$ and $\angle PRQ = \angle B$. PS makes an angle $\frac{\angle P}{3}$ with PQ and RS makes an angle $\frac{\angle SRT}{5}$ with RQ. Then the measure of $\angle S$ is ________



30) In a two-digit positive integer, the units digit is one less than the tens digit. The product of one less than the units digit and one more than the tens digit is 40. The number of such two-digit integers is __________

End of question paper.